Problem 3: The Value of One-Page Innovation

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Salary Determination Model

- $\log\left(\frac{s_{it}}{s_{it-1}}\right) = \gamma_1 \log\left(\frac{Y_{it}}{Y_{it-1}}\right) + \gamma_0 + \gamma_2 ru_{it} + \gamma_3 sex_i + \mu_{it}$

- Check the endogenity problem using Hausman test firstly for this panel data model.

- HTIV method to estimate the model

Estimation Results

	Chisquare	P-Value		
Hausman Test Results:	25.0307	< 3.67e-6		
	Estimate	Std. Error	t-value	Pr(>|t|)
(Intercept)	0.0166487***	0.000786	21.1761	< 2.2e-16
Reputation	0.0524396***	0.001585	33.0862	< 2.2e-16
sex	-0.0057325***	0.001513	-3.7877	0.000152
Employment Status	0.0045888***	0.001261	3.6391	0.000274

Representative Individual

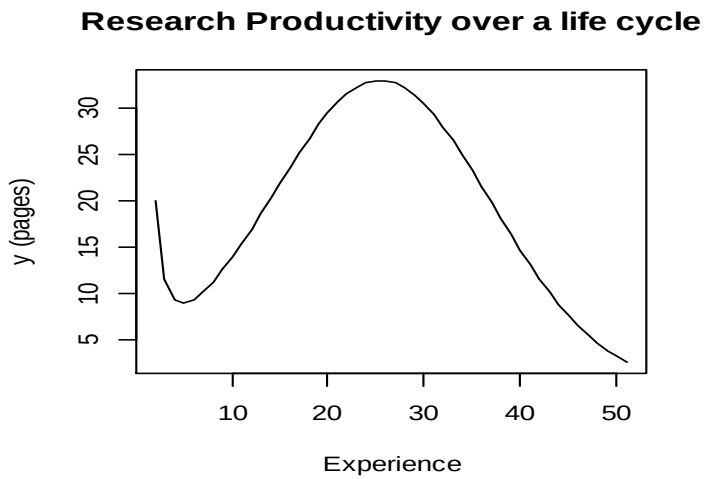
- I choose Taylor Hughes, a male assistant professor in Department of Physics UIUC graduating from Stanford University at 2010, as my representative.

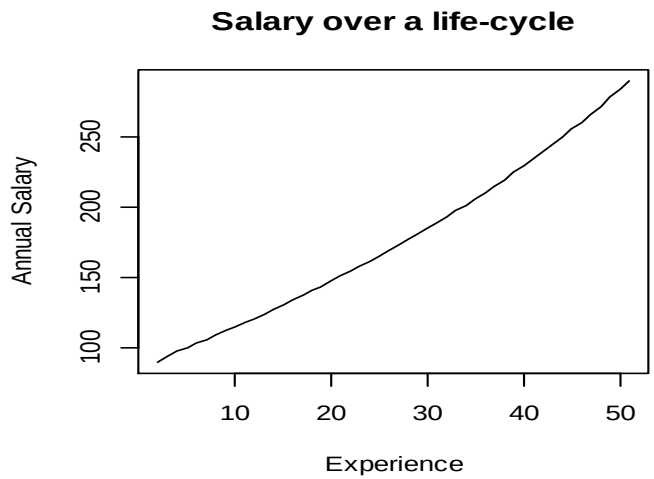
	Life-cycle	Current year	experinece	rank of PhD	sex	employment	Publication Pages	Salary
Taylor Hughes	50	2011	1	4	male	AP in UIUC	23	$91,400

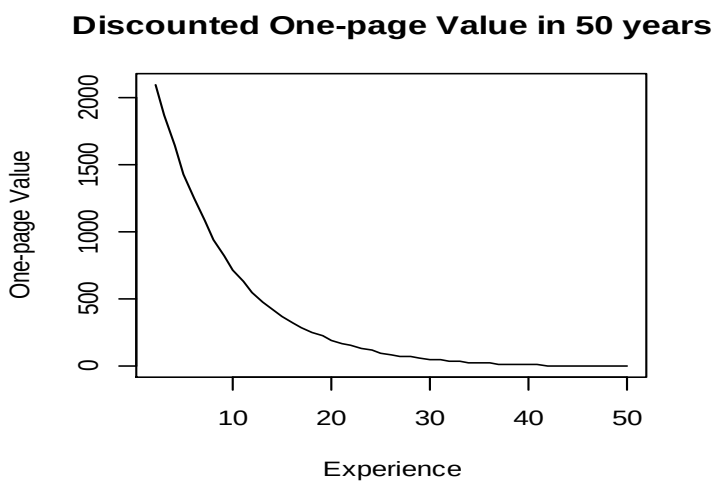
Simulation Procedures

- 1. Given initial conditions, we can predict Taylor's research productivity all over 50 years.
- 2. Using a depreciation rate of 8%, we calculate the discounted cumulative page equivalents each year
- 3. Then we can obtain the annual salary based on the salary determination model and discount it to present value using an inflation rate of 3%
- 4. Choosing different time points, we calculate value of one-page innovation

Prediction Results







Problems

- The salary increases in a almost linear form over years. But there should be salary ceilings in most universities.
- Causality between salary and research productivity

Causality

Beginning of his academia life

- Tenure
- Clark Medal
- Nobel Prize

Research Productivity determines salary

End of his academia life

- Already get tenure
- Love golf
- Realize that one-page innovation just worth $50
- Spend most time on golf rather than research to maximize his utility
- He is rational and salary determines productivity